Lecture 9

Ribbon categories, Temperley–Lieb and $\mathrm{SU}(2)_k$

Objectives. *Lecture 7 found the topological spin as a closed curl, and Lecture 8 the matrix $c = A^{-1}B_q$ of the braiding on two spins. This lecture puts the two together. A twist makes a braided category draw ribbons, not strings, and then every framed knot has a value. On tensor powers of one spin the calculus is the Temperley–Lieb algebra, the value is the Kauffman bracket, and its projectors are the higher spins. At a root of unity one projector vanishes in every closed picture, and discarding it leaves a finite category: $\mathrm{SU}(2)_k$, the source of the Ising-like and Fibonacci-like anyons of Lecture 10.*

By the end of this lecture you will be able to:

- Define a twist and a ribbon category, and show that the curl of Lecture 7 satisfies the twist axiom.
- Compute with the Temperley–Lieb skein relation, and prove that the Kauffman bracket is invariant under Reidemeister II and changes by $-A^{\pm 3}$ under Reidemeister I.
- Construct the Jones–Wenzl projector p_2 and compute traces and twists of coloured strands.
- Explain how $\mathrm{SU}(2)_k$ arises at $A = e^{i\pi/(2(k+2))}$ by discarding negligible morphisms and flipping the pivotal structure.

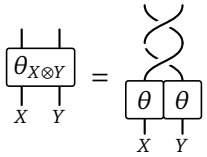
9.1 Twists and ribbon categories

Definition 9.1 (Twist, ribbon category). A *twist* on a braided rigid category is a natural isomorphism $\theta_X : X \to X$ with $\theta_1 = \mathrm{id}$ and

$$\theta_{X\otimes Y} = c_{Y,X}\, c_{X,Y}\, (\theta_X \otimes \theta_Y).$$

It is a *ribbon structure* if moreover $\theta_{X^*} = (\theta_X)^*$ (EGNO Def. 8.10.1; Kassel Def. XIV.3.2).

The axiom is the picture of a ribbon, or a belt: a full turn of a belt holding two strands turns each strand and also wraps them once around each other,



(9.1)

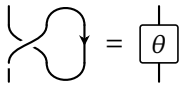
For a simple object θ_a is a number, and $\theta_{a\otimes b}$ acts on the channel c by θ_c; so the axiom is the ribbon relation $R_c^{ab}R_c^{ba} = \theta_c/(\theta_a\theta_b)$ that Lecture 7 cited.

Theorem 9.2 (The curl is a twist · *proved given coherence*)**.** *In a braided pivotal category, the curl of Lecture 7, the right partial trace of $c_{X,X}$, is a twist. In a fusion category with its canonical spherical structure, its value on a simple object is the topological spin θ_a of Theorem 7.7.*

Proof. Naturality: boxes slide through crossings, cups and caps. Invertibility: the curl drawn with the inverse crossing and the cap on the other side is its inverse, by RII and a snake. For $X \otimes Y$, draw the curl of the pair as a belt: the crossing $c_{X\otimes Y, X\otimes Y}$ is four crossings by the hexagons, and closing the right pair of strands with the nested cup and cap of Proposition 4.3 leaves a curl on each strand and the two crossings of $c_{Y,X}c_{X,Y}$ between them, which is the axiom. The value on a simple object is Theorem 7.7, which assumes the canonical structure. ∎

In another pivotal structure the curl can differ by a sign (§ 9.5). In a braided fusion category the curl is a ribbon structure exactly when the pivotal structure is spherical (EGNO Prop. 8.10.12); beyond fusion categories this can fail.

So framing matters. A closed curl is not a straight strand, as it would be in a plane drawing of a knot (Reidemeister I); with a twist, *framed* RI holds:



(9.2)

The strands of a ribbon category are thin bands lying flat in the page (blackboard framing), and a curl puts a full twist into the band.

9.2 Tangles and the Reshetikhin–Turaev functor

A *framed tangle* is a piece of a framed oriented link in a slab, with endpoints on the bottom and top; up to isotopy, tangles form a ribbon category under stacking and juxtaposition.

Theorem 9.3 (Shum; Reshetikhin–Turaev · *cited*)**.** *Framed oriented tangles form the free ribbon category on one object: for every object V of a strict ribbon category C there is a unique strict ribbon functor from tangles to C sending an upward strand to V (Kassel Thm. XIV.5.1).*

A framed link is a tangle from nothing to nothing, so its image is an endomorphism of 1, a number when $\mathrm{End}(1) = \mathbb{C}$, as in a fusion category: a link invariant for every object of such a ribbon category. The pictures of these lectures are honest because of this theorem.

9.3 Temperley–Lieb and the Kauffman bracket

Take $V = V_1$ of $U_q(\mathfrak{sl}_2)$, $q = A^2$, with the cup $\upsilon = e_1 \otimes e_2 - q\, e_2 \otimes e_1$ of Lecture 8. The cap $u\colon V \otimes V \to \mathbb{C}$ is given by

$$u(e_1 \otimes e_1) = u(e_2 \otimes e_2) = 0,$$
$$u(e_1 \otimes e_2) = -q^{-1}, \qquad u(e_2 \otimes e_1) = 1.$$

Both snakes are straight, as in Lecture 4. The contraction $u(\upsilon) = -q^{-1} - q$ gives the closed loop:



$$= \delta = -A^2 - A^{-2} = -[2]_q, \tag{9.3}$$

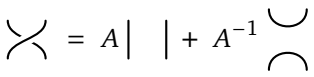
where $[n]_q = (q^n - q^{-n})/(q - q^{-1})$; at $A = \pm 1$ it is the Penrose loop -2 of Lecture 4. Write $U = v \circ u$, a cap followed by a cup. Then $U^2 = \delta U$, and on three strands $U_1U_2U_1 = U_1$, $U_2U_1U_2 = U_2$: the *Temperley–Lieb* relations. The *Temperley–Lieb algebra* $\mathrm{TL}_n(\delta)$ has generators $U_1, \dots, U_{n-1}$, these relations, and $U_iU_j = U_jU_i$ for $|i - j| \geq 2$; by Kauffman's theorem it is the algebra of the diagrams $n \to n$ below (Kauffman–Lins p. 9).

Definition 9.4 (The Temperley–Lieb category)**.** $\mathrm{TL}(\delta)$ has objects $n = 0, 1, 2, \dots$ and as morphisms $n \to m$ the linear combinations of planar diagrams of non-crossing strands joining n points below to m above; composition stacks diagrams and replaces each closed loop by δ.

Theorem 9.5 (Temperley–Lieb acts on spin $\frac{1}{2}$)**.** *With $\delta = -[2]_q$, $U_i \mapsto \mathrm{id} \otimes U \otimes \mathrm{id}$ (acting on strands $i, i+1$) defines a representation of $\mathrm{TL}_n(\delta)$ on $V_1^{\otimes n}$ by U_q-module maps.*

Proof. $U = v \circ u = B_q - q$ (multiply out) is a module map because $B_q = Ac$ is (Lecture 8). Far-apart generators act on different factors and commute; the other relations involve three adjacent strands, where they are the identities above (the *Quantum Computing* survey, §5). ∎

The braiding is then a combination of planar diagrams: from $c = A^{-1}B_q$ and $B_q = q + U$, the *skein relation*



(9.4)

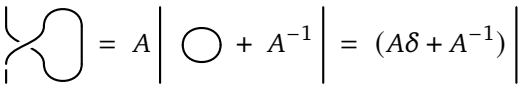
and the inverse crossing exchanges A and A^{-1}. This is the Kauffman bracket's rule as in Simon (Fig. 2.3) and Kauffman–Lins (p. 5); many references, and mirror images, exchange A and A^{-1}.

Theorem 9.6 (The Kauffman bracket · *sketched; proved for RI and RII*)**.** *Evaluating a link diagram by (9.4) and (9.3), the value $\langle L \rangle$ is unchanged by Reidemeister II and III, and a curl multiplies it by $-A^3$, or $-A^{-3}$ for the opposite curl.*

Proof. RII: a crossing followed by its inverse is

$$\begin{aligned}(A + A^{-1}U)(A^{-1} + AU) \\ = 1 + (A^2 + A^{-2})U + U^2 \\ = 1 + (A^2 + A^{-2} + \delta)U = 1.\end{aligned}$$

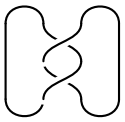
RI: expanding the crossing of a curl gives



and $A\delta + A^{-1} = -A^3$. RIII is the braid relation of Theorem 7.4 for $c = A + A^{-1}U$, which follows from the TL relations alone (Sheet 9). ∎

So $\langle L \rangle$ is an invariant of framed links, and the factor $-A^3$ is the twist θ_{V_1}. For an oriented link, the *writhe* $w(L)$ is the sum of crossing signs. Our /-over crossing of two upward strands has sign +1; reversing one strand reverses the sign. Let O denote the unknot. Then

$f_L(A) = (-A^3)^{-w(L)}\langle L\rangle/\langle O\rangle$ is an invariant of unframed links, and at $A = t^{-1/4}$ it is the *Jones polynomial* $V_L(t)$ (Kauffman–Lins Thm. 1). For the Hopf link with two positive crossings,



$$= \delta\left(-A^4 - A^{-4}\right). \tag{9.5}$$

Indeed, squaring $c = A + A^{-1}U$ and using $U^2 = \delta U$ gives

$$c^2 = A^2 + (2 + A^{-2}\delta)U.$$

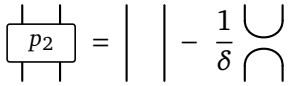
Closing the identity gives two loops. Closing U gives one. Hence

$$\begin{aligned}\langle L\rangle &= A^2\delta^2 + (2 + A^{-2}\delta)\delta \\ &= \delta(-A^4 - A^{-4}).\end{aligned}$$

Thus $\langle L\rangle/\langle O\rangle = -A^4 - A^{-4}$. Sheet 9 does the trefoil.

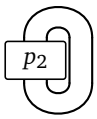
9.4 p_2 and coloured strands

Spin 1 sits inside $V_1 \otimes V_1$ as the image of a projection that kills the cup:



(9.6)

Then $p_2^2 = p_2$ and $Up_2 = 0 = p_2U$, whenever $\delta \neq 0$. Its trace closes both strands:



$$= \delta^2 - \frac{\delta}{\delta} = \delta^2 - 1 = [3]_q, \tag{9.7}$$

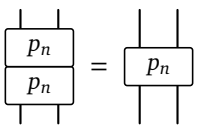
the dimension of spin 1, deformed; at $q = 1$, p_2 is the symmetriser $\frac{1}{2}(1 + P)$ of the triplet, the symmetric part of $\mathbb{C}^2 \otimes \mathbb{C}^2$ (*Symmetries* Ex. 14.13).

Theorem 9.7 (Jones–Wenzl projectors · *cited; proved for p_2*)**.** *On $n+1$ strands, starting from $p_1 = \mathrm{id}$,*

$$p_{n+1} = p_n \otimes 1 + \frac{[n]_q}{[n+1]_q}\,(p_n \otimes 1)\,U_n\,(p_n \otimes 1)$$

defines idempotents killed by every U_i, for as long as $[m]_q \neq 0$ for $m \leq n+1$ (Kauffman–Lins Def. 1 and Lemma 1; for p_2 it is (9.6)). Their traces are $\operatorname{tr} p_n = (-1)^n[n+1]_q$ *(ibid., p. 13).*

In pictures, a projector absorbs a second copy of itself,



(9.8)

and is killed by any cap across two of its strands:

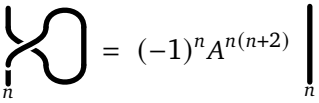


$$= 0 \tag{9.9}$$

A pair (n, p_n), n strands carrying the idempotent p_n, is a *coloured strand*, drawn as one strand labelled n. It is an object of the *idempotent completion*: objects (X, p) with $p^2 = p$, morphisms $f\colon X \to Y$ with $f = p_Y f p_X$. Adding formal direct sums of such pairs gives the *additive* idempotent completion. This is how spin $n/2$ enters the diagram calculus. A coloured strand has a twist, from the full twist of its n strands,

$$\theta_n = (-1)^n A^{n(n+2)}$$

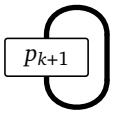
(Kauffman–Lins Prop. 6), and $\theta_1 = -A^3$. In pictures, a curl on a coloured strand is that number:



(9.10)

9.5 Roots of unity, the negligible quotient, the pivotal flip

Fix $k \geq 1$ and $A = e^{i\pi/(2(k+2))}$, so $q = A^2 = e^{i\pi/(k+2)}$. Then $[k+2]_q = 0$ while $[m]_q \neq 0$ for $m \leq k+1$: the recursion reaches p_{k+1}, and $\operatorname{tr} p_{k+1} = (-1)^{k+1}[k+2]_q = 0$:



(9.11)

Definition 9.8 (Negligible). A morphism $f\colon X \to Y$ is *negligible* if $\operatorname{tr}(gf) = 0$ for every $g\colon Y \to X$.

The negligible morphisms form an ideal: closed under composition and $\otimes$ with anything (EGNO Exer. 8.18.9(ii)). By Kauffman–Lins (Lemma 5), p_{k+1} closes to 0 in every network: it is negligible.

Theorem 9.9 ($\mathrm{SU}(2)_k$ · *cited*). *The additive idempotent completion of* $\mathrm{TL}(\delta)$ *at* $A = e^{i\pi/(2(k+2))}$*, modulo negligible morphisms, is a semisimple ribbon category with simple objects the coloured strands* $0, 1, \ldots, k$*, fusing by the truncated Clebsch–Gordan rule of Lecture 5 (Kauffman–Lins Lemmas 4–5; Simon Ch. 22, Tables 22.1–22.3). As a tensor category it is EGNO's* $C_k(q)$ *(Ex. 8.18.5): the quotient of the tilting* U_q*-modules by the negligible ones (Bakalov–Kirillov Thm. 3.3.20). In the pivotal structure of* TL*,*

$$d_n = (-1)^n [n+1]_q,$$
$$\theta_n = (-1)^n A^{n(n+2)}.$$

These d_n are negative for odd n: the Penrose sign of Lecture 4, now on every half-integer spin. The cure is the one Lecture 4 found, a different pivotal structure. Multiplying the pivotal structure by $(-1)^n$ on the colour n is allowed, since $(-1)^n$ is a monoidal function of n (the truncated rule preserves parity); it multiplies d_n and θ_n by $(-1)^n$ (Bakalov–Kirillov Ex. 2.2.8; Simon (15.1)). With $j = n/2$:

$$d_j = [2j+1]_q > 0,$$
$$[2j+1]_q = \frac{\sin\big((2j+1)\pi/(k+2)\big)}{\sin\big(\pi/(k+2)\big)},$$
$$\theta_j = e^{2\pi i\, j(j+1)/(k+2)}.$$

At $k = 2$ the simple objects are 0, $\frac{1}{2}$ and 1, the coloured strands $0, 1, 2$; the truncated rule gives $\frac{1}{2} \otimes \frac{1}{2} \cong 0 \oplus 1$, $\frac{1}{2} \otimes 1 \cong \frac{1}{2}$ (since p_3, which would carry spin $\frac{3}{2}$, is negligible) and $1 \otimes 1 \cong 0$. This is the unitary $\mathrm{SU}(2)_k$ of Lecture 5, with the dimensions found there as Frobenius–Perron dimensions (Bonderson §5.4). At $k = 1$, $\theta_{1/2} = -A^3 = -i$ in the TL structure and $+i$ after the flip: the semion of Lecture 7. At $k = 2$, $d_{1/2} = \sqrt{2}$ and $\theta_{1/2} = e^{3\pi i/8}$, Kitaev's $\nu = 3$: the $\kappa = -1$ category of Lecture 6, not the course's Ising. At $k = 3$, the integer spins 0 and 1 close under fusion, $1 \otimes 1 = 0 \oplus 1$, with $d_1 = [3]_q = \varphi$ and $\theta_1 = e^{4\pi i/5}$: Fibonacci, with the chirality chosen in Lecture 7. The other choice $-A$ of the root multiplies θ_j by $(-1)^{2j}$, and reverses the sign of the braiding $c = A^{-1}B_q$ on two spins.

Why is the quotient semisimple? Each coloured strand $n \le k$ has non-zero trace, so its identity is not negligible and it stays a simple object; the morphisms that would make the category non-semisimple, through p_{k+1} and beyond, are negligible and are set to zero. The traces that survive are the dimensions above.

Summary.

- *Handed on:* a twist satisfies $\theta_{X\otimes Y} = c_{Y,X}c_{X,Y}(\theta_X \otimes \theta_Y)$, and the curl is one (Theorem 9.2); framed tangles form the free ribbon category (Theorem 9.3).
- *Temperley–Lieb:* on $V_1^{\otimes n}$ the crossing is $A \cdot \mathrm{id} + A^{-1} \cdot (\mathrm{cup} \circ \mathrm{cap})$, the loop is $-A^2 - A^{-2}$, and the Kauffman bracket is a framed link invariant (Theorem 9.6).
- *Projectors:* for $\delta \neq 0$, $p_2 = 1 - U/\delta$. The Jones–Wenzl recursion reaches p_n while $[m]_q \neq 0$ for $1 \le m \le n$, with $\mathrm{tr}\, p_n = (-1)^n[n+1]_q$ and $\theta_n = (-1)^n A^{n(n+2)}$ (Theorem 9.7).
- $\mathrm{SU}(2)_k$: at $A = e^{i\pi/(2(k+2))}$, p_{k+1} is negligible; the quotient, with the pivotal flip, has $d_j = [2j+1]_q$ and $\theta_j = e^{2\pi i j(j+1)/(k+2)}$ (Theorem 9.9).
- *Next:* Lecture 10 adds the S-matrix and makes these categories modular.

References

Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: twists and ribbon structures, Def. 8.10.1 and Rem. 8.10.3, p. 216; ribbon iff spherical, Prop. 8.10.12, p. 220; the pivotal flip, Rem. 8.10.16, p. 221; the categories $C_k(q)$ and negligible morphisms, Ex. 8.18.5 and Exer. 8.18.9, pp. 235–236 (their twist $q^{-m(m+2)/2}$ is the inverse of ours). Kassel, *Quantum Groups*: framed tangles, §X.8, pp. 270–272; ribbon categories, Def. XIV.3.2, p. 349; the universal ribbon category, Thm. XIV.5.1, p. 360. Bakalov and Kirillov, *Lectures on Tensor Categories and Modular Functors*: ribbon categories, Def. 2.2.1 and Ex. 2.2.8, pp. 33–35; Reshetikhin–Turaev, Thm. 2.3.10, p. 42; $\mathrm{SU}(2)_k$, Thm. 3.3.20, p. 67. Kauffman and Lins, *Temperley–Lieb Recoupling Theory*: the bracket, pp. 5–9 and Thm. 1; the Jones–Wenzl recursion, Def. 1 and Lemma 1, pp. 13–14; the root of unity, Lemmas 4–5, pp. 37–38; twisted projectors, Prop. 6, p. 43. Simon, *Topological Quantum*: the skein relation, Fig. 2.3, p. 4; twists, Ch. 15, pp. 199–200; Temperley–Lieb–Jones, Ch. 22, pp. 297–310, and the theories at $A = \pm e^{i\pi/(2(k+2))}$, Tables 22.1–22.3. Bonderson, PhD thesis, §5.4, p. 81.